

Aperture — Product Requirements Document (PRD)

Version: v1.0

Status: Draft (Foundation Specification)

Project Codename: Aperture

1. Executive Summary

Aperture is a next-generation cinematic discovery platform that combines the strengths of IMDb, Letterboxd, Trakt, and modern AI-powered recommendation systems into a single cohesive experience.

The platform is not intended to be another movie database or review website.

Its purpose is to become a **cinematic identity platform** that helps users discover, discuss, organize, and experience films and television through intelligent recommendations, beautiful design, and meaningful community interactions.

The project is also intentionally designed to demonstrate production-grade software engineering practices and system design suitable for technical interviews and long-term scalability.

2. Vision

Create the internet's best platform for discovering and discussing cinema by combining:

- Beautiful editorial design
- Rich metadata
- Intelligent discovery
- Community-driven discussion
- AI-powered personalization
- Streaming availability
- Modern engineering

Aperture should feel like a premium product rather than a database.

3. Mission

Help people discover stories they will genuinely love instead of endlessly scrolling through catalogs.

The platform should understand *taste*, not simply viewing history.

4. Product Philosophy

Every major feature should satisfy one or more of these principles:

Discover

Help users find content they would not have found otherwise.

Organize

Allow users to build a permanent cinematic archive.

Express

Allow users to communicate their taste and opinions.

Connect

Create meaningful interactions between users with similar interests.

Learn

Teach users more about cinema through metadata, recommendations, and community.

5. Core Product Pillars

Pillar 1 — Discovery

Discovery is the primary differentiator.

The platform should prioritize:

- Similarity-based recommendations
 - Mood-based discovery
 - Semantic recommendations
 - Editorial collections
 - Natural language discovery
 - Hidden gems
 - Trending content
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Pillar 2 — Community

Community should feel authentic rather than engagement-driven.

Core ideas:

- Reviews
 - Ratings
 - Lists
 - Activity feeds
 - Movie Spaces
 - Following users
 - Discussion
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Pillar 3 — Personal Identity

Each user profile should become a representation of their cinematic taste.

The profile evolves over time based on:

- Ratings
 - Reviews
 - Favorites
 - Watch history
 - Lists
 - Recommendations
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Pillar 4 — Intelligence

AI should improve the experience without becoming the product.

It should support:

- Better recommendations
 - Better search
 - Better categorization
 - Better personalization
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Pillar 5 — Beautiful Design

The interface should feel premium.

Design goals:

- Minimal
 - Cinematic
 - Editorial
 - Responsive
 - Accessible
 - Motion-driven but subtle
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6. Branding

Name

Aperture

Meaning

An aperture controls how light enters a camera.

Metaphorically:

Aperture is the lens through which users discover cinema.

Brand Personality

- Elegant
 - Modern
 - Intelligent
 - Cinematic
 - Artistic
 - Calm
 - Sophisticated
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Visual Inspiration

- A24
- MUBI
- Apple TV+
- Criterion Collection
- Letterboxd (community)

- Premium editorial magazines
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7. Target Audience

Primary

Film enthusiasts who:

- Track movies
 - Write reviews
 - Create lists
 - Enjoy discussing cinema
 - Care about cinematography and storytelling
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Secondary

General streaming users who:

- Want better recommendations
 - Want to know where to watch
 - Keep personal watchlists
 - Occasionally review content
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8. Metadata Strategy

Primary metadata provider:

The Movie Database (TMDb)

Reasons:

- Excellent movie support
- Excellent TV support
- High-quality artwork
- Rich metadata
- Watch provider information
- Mature API ecosystem

Aperture will maintain its own canonical data model so future metadata sources can be added without changing the application architecture.

9. Signature Features

Emotional Discovery Engine

Recommend movies based on:

- mood
- atmosphere
- pacing
- emotional tone
- visual style
- themes

rather than only genres.

Cinematic DNA

Each user receives a continuously evolving cinematic profile based on their interactions.

Potential outputs include:

- Taste archetypes
 - Genre affinity
 - Director affinity
 - Emotional preference map
 - Viewing evolution
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AI Taste Matching

Match users based on similarity of cinematic preferences instead of follower count or popularity.

Movie Spaces

Communities centered around:

- directors
- genres
- themes
- aesthetics
- moods
- franchises

10. Core Feature Catalogue

Authentication

- Email/password
 - Google OAuth
 - Session management
 - Secure authentication
 - Profile management
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User Profiles

- Avatar
 - Biography
 - Favorites
 - Reviews
 - Lists
 - Followers
 - Following
 - Activity history
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Movies & TV

- Detail pages
 - Cast & crew
 - Trailers
 - Images
 - Similar titles
 - Streaming availability
 - Collections
 - Related works
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Reviews

- Ratings
 - Long-form reviews
 - Spoiler support
 - Likes
 - Comments
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Lists

- Watchlists
 - Custom collections
 - Ranked lists
 - Public/private visibility
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Search

- Movies
 - TV shows
 - People
 - Users
 - Lists
 - Spaces
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Discovery

- Trending
 - Popular
 - Hidden gems
 - Semantic recommendations
 - Editorial collections
 - Mood exploration
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Streaming

- Provider availability
 - Region-aware availability
 - Subscription filtering
 - Rent/buy options
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Community

- Follow users
 - Activity feed
 - Notifications
 - Comments
 - Reactions
 - Movie Spaces
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11. Non-Functional Requirements

The platform should prioritize:

Performance

- Fast page loads
 - Optimized search
 - Efficient caching
 - Responsive UI
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Reliability

- Stable APIs
 - Graceful error handling
 - Background synchronization
 - Monitoring
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Scalability

Architecture should evolve naturally from:

Modular Monolith

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Selective Service Extraction

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Distributed Architecture (only if necessary)

Security

- Authentication
 - Authorization
 - Input validation
 - Rate limiting
 - Secure secrets management
 - OWASP best practices
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Accessibility

Target WCAG AA compliance through:

- Keyboard navigation
 - Screen reader support
 - Color contrast
 - Motion preferences
 - Semantic HTML
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12. Success Metrics

Product metrics:

- Daily active users
- Reviews written
- Lists created
- Recommendation click-through rate
- Average session duration
- Watchlist additions
- Community engagement

Engineering metrics:

- API latency
 - Search latency
 - Cache hit ratio
 - Error rate
 - Test coverage
 - Deployment frequency
 - Mean time to recovery (MTTR)
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13. Long-Term Vision

Aperture should evolve from:

Movie database

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Movie discovery platform

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Social cinematic community

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AI-powered taste platform

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Comprehensive cultural discovery ecosystem

Potential future expansion:

- Books
 - Games
 - Anime
 - Documentaries
 - Podcasts
 - Music
 - Visual arts
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14. Product Principles

Every new feature should answer "yes" to at least one of the following:

- Does it improve discovery?
- Does it strengthen community?
- Does it enhance cinematic identity?
- Does it simplify organization?
- Does it make the experience more beautiful?
- Does it provide meaningful insight?

If the answer is "no" to all of the above, the feature should be reconsidered.

15. Out of Scope (Initial Releases)

The following are intentionally deferred to later phases:

- Mobile applications
- Offline viewing support
- Video hosting

- Live streaming
- Real-time watch parties
- Creator monetization
- Marketplace features
- Multi-tenant deployments

These may be explored once the core platform achieves product maturity.

16. Document Relationships

This PRD defines **what** Aperture is.

The following documents derive directly from it:

- Development Roadmap
- High-Level Design
- Low-Level Design
- Database Design
- AI System Design
- Infrastructure & DevOps
- Engineering Handbook
- Architecture Decision Records

Any future feature or architectural change should first be evaluated against the vision and principles established in this document.